

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)		Vertical: Decreasing (accept deceleration), then increasing (accept acceleration) / <u>changes</u> at 9.81 m s^{-2} (1) Horizontal: Constant (1) Reason: Gravity acts vertically or no forces act horizontally (1)	3			3		
	(b)	(i)	$0.15 [\text{m s}^{-1}]$ i.e. $\frac{(1000)}{(110 \times 60)}$ or 0.54 km/h		1		1		
		(ii)	Correct substitution into $x = ut + \frac{1}{2}at^2$ Ignore sign convention e.g. $1000 = \frac{1}{2} \times a \times (55 \times 60)^2$ (1) At least one mathematical step shown leading to $a = 0.00018 [\text{m s}^{-2}]$ e.g. $a = \frac{2000}{1.09 \times 10^7}$ (1) Alternative: u_{vertical} calculated from $x = \frac{1}{2}(u + v)t$ i.e. $u = 0.606 \text{ m s}^{-1}$ (1) Substitution into: $a = (v - u)/t$ to show $a = 0.00018 [\text{m s}^{-2}]$ (1)	1	1		2	2	
		(iii)	Correct substitution into $v = u + at$ or $v^2 = u^2 + 2ax$ e.g. $0 = u - 0.00018 (55 \times 60)$ or $0 = u^2 - 2 \times 0.00018 \times 1000$ (1) ecf [accept use of 0.0002 m s^{-2}] e.g. $u = 0.61 [\text{m s}^{-1}]$ (1) e.g. $\frac{0.61}{0.88} \times 100\%$ seen (1) Accept $67\% - 75\%$ Alternative for final mark: $60\% \text{ of } 0.88 \text{ m s}^{-1} = 0.53 \text{ m s}^{-1}$ therefore: $0.61 > 0.53$	1	1 1		3	3	

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	(c)		For • Job creation • Cost/year reasonable • Generate interest in science • New technologies developed e.g. renewable • Improve understanding of origin of life on earth	Against • Funding could have been used to address earth based issues. • Little impact on society • Costs outweigh discoveries • Risky mission it might have failed			3	3		
			3 statements given must expand on bullet points in question (3) No mark for agreeing or disagreeing							
			Question 6 total		5	4	3	12	5	0